Chapter 6. Simple Linear Regression

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Simple linear regression, the basis of the regression estimator (Ch. 7), compresses n-dimensional data into $(\hat{\beta}_0, \hat{\beta}_1)$: data = model structure + error.

1 The Model

Definition 6.1

The simple linear regression model with a single regressor x is

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad i = 1, \ldots, n,$$

with uncorrelated errors of mean 0 and variance σ^2; x is fixed, y random; β_0, β_1 are the **regression coefficients**.

Then

$$\mathrm{E}(y \mid x) = \beta_0 + \beta_1 x = f(x), \qquad \mathrm{Var}(y \mid x) = \mathrm{Var}(\varepsilon) = \sigma^2,$$

so the true regression line is the line of mean responses. The goals are to estimate β_0, β_1 and to test $H_0 : \beta_1 = 0$.

2 Least-Squares Estimation

Motivation. With an intercept, a regression line replaces the ratio model; we review least squares.

The least-squares (LS) estimators minimize

$$S(\beta_0,\beta_1)=\sum_{i=1}^{n}(y_i-\beta_0-\beta_1x_i)^2=\sum_{i=1}^{n}\varepsilon_i^2.$$

Setting $\partial S/\partial\beta_0=\partial S/\partial\beta_1=0$ gives the **normal equations**

$$n\hat\beta_0+\hat\beta_1\sum x_i=\sum y_i,\qquad \hat\beta_0\sum x_i+\hat\beta_1\sum x_i^2=\sum x_iy_i.$$

Theorem 6.2: LS estimators

$$\hat\beta_1=\frac{S_{xy}}{S_{xx}}=\frac{\sum(x_i-\bar x)(y_i-\bar y)}{\sum(x_i-\bar x)^2}=\frac{\sum(x_i-\bar x)y_i}{\sum(x_i-\bar x)^2},\qquad \hat\beta_0=\bar y-\hat\beta_1\bar x.$$

The **fitted values** are $\hat y_i=\hat\beta_0+\hat\beta_1x_i=\widehat{\mathrm{E}}(y_i\mid x_i)$ and the **residuals** are $e_i=y_i-\hat y_i$, so $y_i=\hat\beta_0+\hat\beta_1x_i+e_i$; residuals are the basis of model-adequacy checks (residual analysis).

Example 6.3: Rocket propellant data, $n=20$

$y=$ shear strength, $x=$ age of propellant. From $\bar x=13.3625$, $\bar y=2131.358$, $S_{xx}=1106.559$, $S_{xy}=-41112.65$:

$$\hat\beta_1=\frac{-41112.65}{1106.559}=-37.15,\qquad \hat\beta_0=2131.358+37.15\times13.3625=2627.82,$$

so $\hat y=2627.82-37.15x$. R output: intercept 2627.822 (SE 44.184), slope -37.154 (SE 2.889, $t=-12.86$, $p=1.64\times10^{-10}$).

2.1 Properties of the estimators

Theorem 6.4

$\mathrm{E}(\hat\beta_1)=\beta_1$, $\mathrm{E}(\hat\beta_0)=\beta_0$, and

$$\mathrm{Var}(\hat\beta_1)=\frac{\sigma^2}{S_{xx}},\qquad \mathrm{Var}(\hat\beta_0)=\sigma^2\Big(\frac1n+\frac{\bar x^2}{S_{xx}}\Big),\qquad \mathrm{Cov}(\hat\beta_0,\hat\beta_1)=-\bar x\frac{\sigma^2}{S_{xx}}.$$

Derivation: proof

Write $\hat\beta_1=\sum c_iy_i$ with $c_i=(x_i-\bar x)/S_{xx}$. Then $\sum c_i=0$ and $\sum c_ix_i=\sum(x_i-\bar x)x_i/S_{xx}=1$, so

$$\mathrm{E}(\hat\beta_1)=\sum c_i(\beta_0+\beta_1x_i)=\beta_0\sum c_i+\beta_1\sum c_ix_i=\beta_1,$$

$$\mathrm{Var}(\hat\beta_1)=\sigma^2\sum c_i^2=\sigma^2\frac{S_{xx}}{S_{xx}^2}=\frac{\sigma^2}{S_{xx}}.$$

Since $\mathrm{E}(\bar y)=\beta_0+\beta_1\bar x$, $\mathrm{E}(\hat\beta_0)=\beta_0+\beta_1\bar x-\bar x\beta_1=\beta_0$. With $\bar y=\sum d_iy_i$, $d_i=1/n$,

$$\mathrm{Cov}(\bar y,\hat\beta_1)=\sigma^2\sum d_ic_i=\frac{\sigma^2}{n}\sum c_i=0,$$

hence

$$\mathrm{Var}(\hat\beta_0) = \mathrm{Var}(\bar y) + \bar x^2 \mathrm{Var}(\hat\beta_1) - 2\bar x \mathrm{Cov}(\bar y, \hat\beta_1) = \sigma^2\Big(\frac{1}{n} + \frac{\bar x^2}{S_{xx}}\Big),$$

$$\mathrm{Cov}(\hat\beta_0, \hat\beta_1) = \mathrm{Cov}(\bar y, \hat\beta_1) - \bar x \mathrm{Var}(\hat\beta_1) = -\bar x \sigma^2 / S_{xx}.$$

2.2 Properties of the fit

Theorem 6.5

(1) $\sum e_i = 0$, equivalently $\sum y_i = \sum \hat y_i$;

(2) the fitted line passes through $(\bar x, \bar y)$;

(3) $\sum x_i e_i = 0$;

(4) $\sum \hat y_i e_i = 0$.

Proof. (1) and (3) are the two normal equations: $\sum(y_i - \hat\beta_0 - \hat\beta_1 x_i) = 0$ and $\sum x_i(y_i - \hat\beta_0 - \hat\beta_1 x_i) = 0$. (2): at $x = \bar x$, $\hat y = \hat\beta_0 + \hat\beta_1 \bar x = \bar y$. (4): $\sum \hat y_i e_i = \hat\beta_0 \sum e_i + \hat\beta_1 \sum x_i e_i = 0$. ■

2.3 Centered model

Rewriting

$$y_i = (\beta_0 + \beta_1 \bar x) + \beta_1(x_i - \bar x) + \varepsilon_i = \beta_0' + \beta_1(x_i - \bar x) + \varepsilon_i$$

gives $\hat\beta_0' = \bar y$ and $\hat\beta_1 = S_{xy}/S_{xx}$, and now $\mathrm{Cov}(\hat\beta_0', \hat\beta_1) = \mathrm{Cov}(\bar y, \hat\beta_1) = 0$: centering makes the estimators uncorrelated.

3 Estimation of σ^2

Motivation. Inference on the slope needs the residual mean square.

Theorem 6.6: Sum-of-squares decomposition

$$\underbrace{\sum (y_i - \bar y)^2}_{SS_T} = \underbrace{\sum (y_i - \hat y_i)^2}_{SS_E} + \underbrace{\sum (\hat y_i - \bar y)^2}_{SS_R}, \qquad SS_E = SS_T - \hat\beta_1 S_{xy}, \quad SS_R = \hat\beta_1 S_{xy}.$$

Proof. Expanding $\sum(y_i - \hat y_i + \hat y_i - \bar y)^2$ leaves the cross term $2\sum e_i(\hat y_i - \bar y) = 2(\sum e_i \hat y_i - \bar y \sum e_i) = 0$. Since $e_i = (y_i - \bar y) - \hat\beta_1(x_i - \bar x)$,

$$SS_E = SS_T + \hat\beta_1^2 S_{xx} - 2\hat\beta_1 S_{xy} = SS_T - \hat\beta_1 S_{xy}.$$

■

SS_E has $n - 2$ degrees of freedom, and the unbiased estimator of σ^2 is

$$\hat\sigma^2 = \frac{SS_E}{n-2} = MS_E \quad \text{(the mean square error)}.$$

$\hat\sigma$ is the standard error of regression; it is a model-dependent estimate of σ.

Example 6.7: Rocket propellant

$SS_T = 1{,}693{,}737.6$ and $SS_E = 1{,}693{,}737.6 - (-37.15)(-41112.65) = 166{,}402.65$, so $\hat{\sigma}^2 = 166402.65/18 = 9244.59$.

4 Hypothesis Testing on the Slope

Motivation. A non-significant slope means the auxiliary variable helps little.

Now assume $\varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$. Then $\hat{\beta}_1 \sim N(\beta_1, \sigma^2/S_{xx})$.

Theorem 6.8: t-test for significance of regression

For $H_0 : \beta_1 = 0$ versus $H_1 : \beta_1 \neq 0$,

$$t_0 = \frac{\hat{\beta}_1}{\text{se}(\hat{\beta}_1)}, \qquad \text{se}(\hat{\beta}_1) = \sqrt{MS_E/S_{xx}},$$

which is t_{n-2} under H_0. Reject H_0 if $|t_0| > t_{\alpha/2,n-2}$ (equivalently, if the P-value $\leq \alpha$). If σ^2 were known, $Z_0 = \hat{\beta}_1/\sqrt{\sigma^2/S_{xx}} \sim N(0,1)$ would be used instead.

Failing to reject H_0 means there is no evidence of a *linear* relationship between x and y.

Example 6.9: Rocket propellant

$t_0 = -37.15/\sqrt{9244.59/1106.56} = -12.85$ and $t_{0.025,18} = 2.101$, so H_0 is rejected; the P-value is 1.64×10^{-10}.

4.1 Analysis of variance

Theorem 6.10: F-test

With degrees of freedom $(n-1) = 1 + (n-2)$,

$$F_0 = \frac{SS_R/1}{SS_E/(n-2)} = \frac{MS_R}{MS_E} \sim F_{1,n-2} \quad \text{under } H_0,$$

and H_0 is rejected if $F_0 > F_{\alpha,1,n-2}$. Since $\text{E}(MS_R) = \sigma^2 + \beta_1^2 S_{xx}$ and $\text{E}(MS_E) = \sigma^2$, F_0 is near 1 under H_0 and large otherwise. In simple regression $F_0 = t_0^2$.

ANOVA gives the same answer as the t-test here, but it is the natural tool for multiple regression.

Example 6.11: Rocket propellant

$SS_R = (-37.15)(-41112.65) = 1{,}527{,}334.95$, $SS_E = 166{,}402.65$,

$$F_0 = \frac{1527334.95/1}{166402.65/18} = 165.21 > F_{0.05,1,18} = 4.41,$$

so H_0 is rejected (R: $F = 165.4$ on 1 and 18 df, $p = 1.64 \times 10^{-10}$).

5 Interval Estimation

5.1 Coefficients

Under normal errors, $(\hat\beta_j - \beta_j)/\mathrm{se}(\hat\beta_j) \sim t_{n-2}$, so

$$\hat\beta_1 \pm t_{\alpha/2,n-2}\,\mathrm{se}(\hat\beta_1), \qquad \hat\beta_0 \pm t_{\alpha/2,n-2}\,\mathrm{se}(\hat\beta_0),$$

$$\mathrm{se}(\hat\beta_1) = \sqrt{\frac{MS_E}{S_{xx}}}, \qquad \mathrm{se}(\hat\beta_0) = \sqrt{MS_E\Big(\frac{1}{n} + \frac{\bar x^2}{S_{xx}}\Big)}.$$

Example 6.12

For the rocket data with $t_{0.025,18} = 2.101$:

$$2534.99 \le \beta_0 \le 2720.65, \qquad -43.22 \le \beta_1 \le -31.08.$$

5.2 Error variance

Since $(n-2)MS_E/\sigma^2 \sim \chi^2_{n-2}$,

$$\frac{(n-2)MS_E}{\chi^2_{\alpha/2,n-2}} \le \sigma^2 \le \frac{(n-2)MS_E}{\chi^2_{1-\alpha/2,n-2}}.$$

5.3 Mean response

For x_0 within the range of the data, the estimate is $\hat\mu_{y|x_0} = \hat\beta_0 + \hat\beta_1 x_0$, with $\mathrm{E}(\hat\mu_{y|x_0}) = \beta_0 + \beta_1 x_0$ and

$$\begin{aligned}\mathrm{Var}(\hat\mu_{y|x_0}) &= \mathrm{Var}(\hat\beta_0) + x_0^2\mathrm{Var}(\hat\beta_1) + 2x_0\mathrm{Cov}(\hat\beta_0,\hat\beta_1)\\ &= \sigma^2\Big(\frac{1}{n} + \frac{\bar x^2}{S_{xx}}\Big) + x_0^2\frac{\sigma^2}{S_{xx}} - 2x_0\bar x\frac{\sigma^2}{S_{xx}} = \sigma^2\Big(\frac{1}{n} + \frac{(x_0-\bar x)^2}{S_{xx}}\Big).\end{aligned}$$

Hence

$$\hat\mu_{y|x_0} \pm t_{\alpha/2,n-2}\sqrt{MS_E\Big(\frac{1}{n} + \frac{(x_0-\bar x)^2}{S_{xx}}\Big)}.$$

At $x_0 = \bar x = 13.3625$ in the rocket data the 95% interval is $2086.2 \le \mathrm{E}(y \mid x_0) \le 2176.6$. The interval is narrowest at $\bar x$ and widens away from it.

5.4 Prediction of a new observation

The point prediction of y_0 at x_0 is $\hat y_0 = \hat\beta_0 + \hat\beta_1 x_0$. Since the new y_0 is independent of the data,

$$\mathrm{E}(\hat y_0 - y_0)^2 = \mathrm{E}[\hat y_0 - \mathrm{E}(y_0)]^2 + \mathrm{E}[y_0 - \mathrm{E}(y_0)]^2 = \sigma^2\Big(1 + \frac{1}{n} + \frac{(x_0-\bar x)^2}{S_{xx}}\Big),$$

and the $100(1-\alpha)\%$ prediction interval is

$$\hat y_0 \pm t_{\alpha/2,n-2}\sqrt{MS_E\Big(1 + \frac{1}{n} + \frac{(x_0-\bar x)^2}{S_{xx}}\Big)}.$$

For the rocket data at $x_0 = 10$: $[2048.32,\ 2464.32]$.

6 Coefficient of Determination

Definition 6.13

$R^2 = \dfrac{SS_R}{SS_T} = 1 - \dfrac{SS_E}{SS_T}$, $0 \le R^2 \le 1$: the proportion of variation in y explained by x.

In simple regression,

$$R^2 = \frac{\hat\beta_1 S_{xy}}{S_{yy}} = \frac{S_{xy}^2}{S_{xx}S_{yy}} = r_{xy}^2, \qquad F_0 = \frac{SS_R/1}{SS_E/(n-2)} = \frac{R^2(n-2)}{1-R^2}.$$

For the rocket data $R^2 = 1527334.95/1693737.6 = 0.9018 = (-0.9497)^2$: 90.18% of the variability in strength is explained by age.

7 Selected exercises

Motivation. Proofs of the unbiasedness of MS_E, the expected regression mean square and LS optimality, and the no-intercept model linking to Ch. 5.

Exercise 6.1 (Expected mean squares)

In the simple linear regression model show that
(a) $\mathrm{E}(SS_R) = \sigma^2 + \beta_1^2 S_{xx}$;
(b) $\mathrm{E}(SS_E) = (n-2)\sigma^2$, so MS_E is unbiased for σ^2;
(c) $F_0 = t_0^2$.

Solution. [Supplement]

(a) Since $SS_R = \hat\beta_1 S_{xy}$ and $S_{xy} = \hat\beta_1 S_{xx}$, $SS_R = \hat\beta_1^2 S_{xx}$. Hence, by the theorem on the properties of the estimators,

$$\mathrm{E}(SS_R) = S_{xx}\big[\mathrm{Var}(\hat\beta_1) + (\mathrm{E}\hat\beta_1)^2\big] = S_{xx}\Big(\frac{\sigma^2}{S_{xx}} + \beta_1^2\Big) = \sigma^2 + \beta_1^2 S_{xx}.$$

(b)

Step 1. Averaging the model, $\bar y = \beta_0 + \beta_1\bar x + \bar\varepsilon$, so $y_i - \bar y = \beta_1(x_i - \bar x) + (\varepsilon_i - \bar\varepsilon)$.

Step 2. Squaring and summing, the cross term $2\beta_1\sum(x_i-\bar x)(\varepsilon_i-\bar\varepsilon)$ has mean zero, and

$$\mathrm{E}\sum(\varepsilon_i - \bar\varepsilon)^2 = \mathrm{E}\Big[\sum\varepsilon_i^2 - n\bar\varepsilon^2\Big] = n\sigma^2 - n\cdot\frac{\sigma^2}{n} = (n-1)\sigma^2,$$

using that the errors are uncorrelated. Hence $\mathrm{E}(SS_T) = \beta_1^2 S_{xx} + (n-1)\sigma^2$.

Step 3. By the sum-of-squares decomposition and (a), $\mathrm{E}(SS_E) = \mathrm{E}(SS_T) - \mathrm{E}(SS_R) = (n-1)\sigma^2 - \sigma^2 = (n-2)\sigma^2$.

(c) $F_0 = \dfrac{\hat\beta_1^2 S_{xx}}{MS_E} = \Big(\dfrac{\hat\beta_1}{\sqrt{MS_E/S_{xx}}}\Big)^2 = t_0^2$. Under H_0, (a) and (b) give $\mathrm{E}(MS_R) = \mathrm{E}(MS_E) = \sigma^2$, which is why F_0 is near 1. ■

Exercise 6.2 (Least squares is best among linear unbiased estimators)

Show that $\hat\beta_1$ has the smallest variance among all estimators of the form $\sum_i a_i y_i$ that are unbiased for β_1 whatever β_0, β_1 are.

Solution. [Supplement]

Step 1. (Constraints.) $\mathrm{E}\sum a_i y_i = \beta_0 \sum a_i + \beta_1 \sum a_i x_i$ equals β_1 for all (β_0, β_1) iff $\sum a_i = 0$ and $\sum a_i x_i = 1$. Its variance is $\sigma^2 \sum a_i^2$.

Step 2. (Decomposition.) Let $c_i = (x_i - \bar{x})/S_{xx}$, so $\hat{\beta}_1 = \sum c_i y_i$ and the c_i satisfy both constraints. Put $d_i = a_i - c_i$; then $\sum d_i = 0$ and $\sum d_i x_i = 0$.

Step 3. (Orthogonality.) $\sum c_i d_i = \frac{1}{S_{xx}}\big(\sum d_i x_i - \bar{x} \sum d_i\big) = 0$, so

$$\mathrm{Var}\Big(\sum a_i y_i\Big) = \sigma^2 \sum (c_i + d_i)^2 = \sigma^2 \sum c_i^2 + \sigma^2 \sum d_i^2 \;\geq\; \sigma^2 \sum c_i^2 = \mathrm{Var}(\hat{\beta}_1),$$

with equality iff all $d_i = 0$, i.e. the estimator is $\hat{\beta}_1$ itself. This is the Gauss–Markov theorem for the slope; normality is not needed. ■

Exercise 6.3 (Regression through the origin)

Let $y_i = \beta x_i + \varepsilon_i$, $i = 1, \ldots, n$, with fixed x_i not all zero and uncorrelated errors of mean 0 and variance σ^2.
(a) Find the LS estimator $\hat{\beta}$, its mean and variance.
(b) With $e_i = y_i - \hat{\beta} x_i$, show that $\sum x_i e_i = 0$ but in general $\sum e_i \neq 0$.
(c) Show that $\mathrm{E}\sum e_i^2 = (n-1)\sigma^2$, so $\hat{\sigma}^2 = \sum e_i^2/(n-1)$ is unbiased.

Solution. [Supplement]

(a) $\frac{d}{d\beta}\sum (y_i - \beta x_i)^2 = -2\sum x_i(y_i - \beta x_i) = 0$ gives $\hat{\beta} = \sum x_i y_i / \sum x_i^2$, the unique minimizer since the second derivative $2\sum x_i^2$ is positive. Then $\mathrm{E}\hat{\beta} = \beta \sum x_i^2 / \sum x_i^2 = \beta$ and $\mathrm{Var}\hat{\beta} = \sigma^2 \sum x_i^2 / (\sum x_i^2)^2 = \sigma^2 / \sum x_i^2$.

(b) The normal equation of (a) is $\sum x_i e_i = 0$. There is no intercept, hence no equation $\sum e_i = 0$: for the data $(x, y) = (1, 1), (2, 1)$, $\hat{\beta} = 3/5$ and $e_1 + e_2 = 0.4 - 0.2 = 0.2 \neq 0$.

(c)

Step 1. By (b), $\sum e_i^2 = \sum e_i y_i - \hat{\beta} \sum e_i x_i = \sum e_i y_i = \sum y_i^2 - \hat{\beta} \sum x_i y_i = \sum y_i^2 - \hat{\beta}^2 \sum x_i^2$.

Step 2. $\mathrm{E}\sum y_i^2 = \sum (\sigma^2 + \beta^2 x_i^2) = n\sigma^2 + \beta^2 \sum x_i^2$, and by (a) $\mathrm{E}(\hat{\beta}^2) \sum x_i^2 = (\sigma^2 / \sum x_i^2 + \beta^2) \sum x_i^2 = \sigma^2 + \beta^2 \sum x_i^2$.

Step 3. Subtracting, $\mathrm{E}\sum e_i^2 = (n-1)\sigma^2$: only one degree of freedom is lost, for the single coefficient.

With constant variance the no-intercept slope is $\sum x_i y_i / \sum x_i^2$; with variance $\propto x_i$ it is $\sum y_i / \sum x_i$ (Exercise 5.1): the ratio estimator is regression through the origin. ■